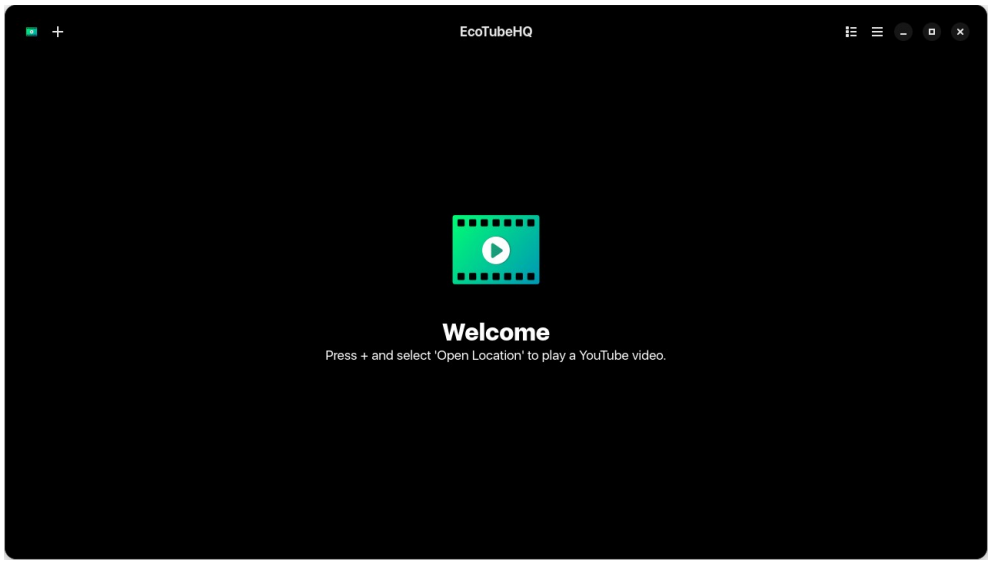
EcoTubeHQ

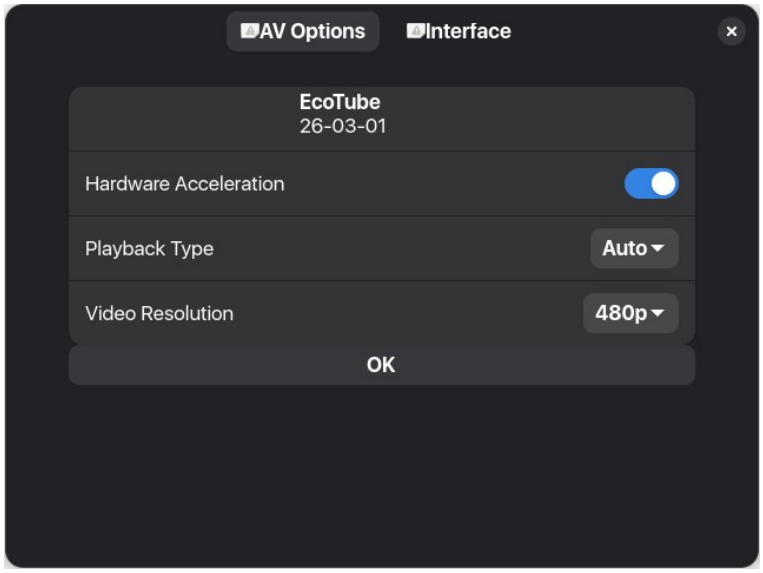
EcoTubeHQ Video Player offers high quality video playback for many DRM-free streaming services whilst using minimal resources. This is achieved through automated video codec selection, high quality video upscaling, hardware acceleration and minimal buffering.



EcoTubeHQ offers AMD FSR [FidelityFX Super Resolution] to produce high quality video output. AMD FSR works with most GPUs and iGPUs [list here].

Player Preferences

Click on the hamburger menu next to the minimize button, then select Preferences. This shows the following…



...here’s what it means:

Hardware Acceleration

On [Default] - Vulkan GPU Compute Acceleration offloads decoding to your GPU for av1, vp9, h.264, and h.265 video codecs, reducing CPU usage. This improves playback on older PCs and extends laptop battery life. Disable if you experience playback problems.

Playback Type

Auto [Default] - **Quality** is the default setting on desktop computers. Laptops use **Quality** when plugged in and automatically switch to **Powersave** on battery power.

> **Quality** - av1 video is upscaled by AMD FSR to optimize video quality [if av1 is not available then vp9 is output, and if vp9 is not available then h.264 is used]. Debanding is also employed to show colors as smooth gradients rather than distinct bands, further improving picture quality [example image here]. The video output quality @720p is roughly equivalent to standard 1080p video output.
>
> **Powersave** - vp9 video uses Nearest Neighbor video upscaling. This gives HQ video output with minimal CPU usage [if vp9 is not available then h.264 is output].

Video Resolution

Supported Video Resolutions: 720p, **480p [Default],** 360p, 240p, 144p

The **Quality** playback mode restricts the maximum video resolution to 720p to minimize data usage, and **Powersave** caps the resolution at 480p to minimize CPU usage.

Note: EcoTubeHQ doesn't offer 60fps because it requires 50%+ additional data compared to 30fps.

Note 2: YouTube restricts some old h.264 videos to three resolutions: 144p, 360p, and 720p. If EcoTubeHQ requests an unavailable h.264 resolution [240p or 480p], it automatically switches to the next highest tier - 240p requests play at 360p, and 480p requests play at 720p.

Note 3: ***EcoTubeHQ automatically updates yt-dlp and FFmpeg to minimize issues.***

> *If the player regularly displays the error* ***'Playback was terminated abnormally. Reason: loading failed'*** *when you try and play a video then there may be a problem with the streaming server or the software may need to update. Simply close the player for a while and try again later.*

Minimal Buffering

EcoTubeHQ uses a 10 second buffer during video playback. In comparison, a conventional buffer for YouTube can use up to 90 seconds of data. I'll use these numbers to explain how EcotubeHQ minimizes buffering, thus reducing wasted data.

Let's say EcoTubeHQ outputs 31 seconds of a video before playback is stopped. This means three 10 second buffers have been downloaded and output, and the fourth was fully downloaded and stopped after one second, thus leaving nine seconds of wasted A/V data in the buffer.

Figure 1: 9 seconds of wasted data [shown in grey]

If conventional buffering is used and playback of the video is stopped after an identical 31 seconds then there is approx 90 seconds of unwatched A/V data in the PC's buffer.

Figure 2: 90 seconds of wasted data [shown in grey - not to scale]

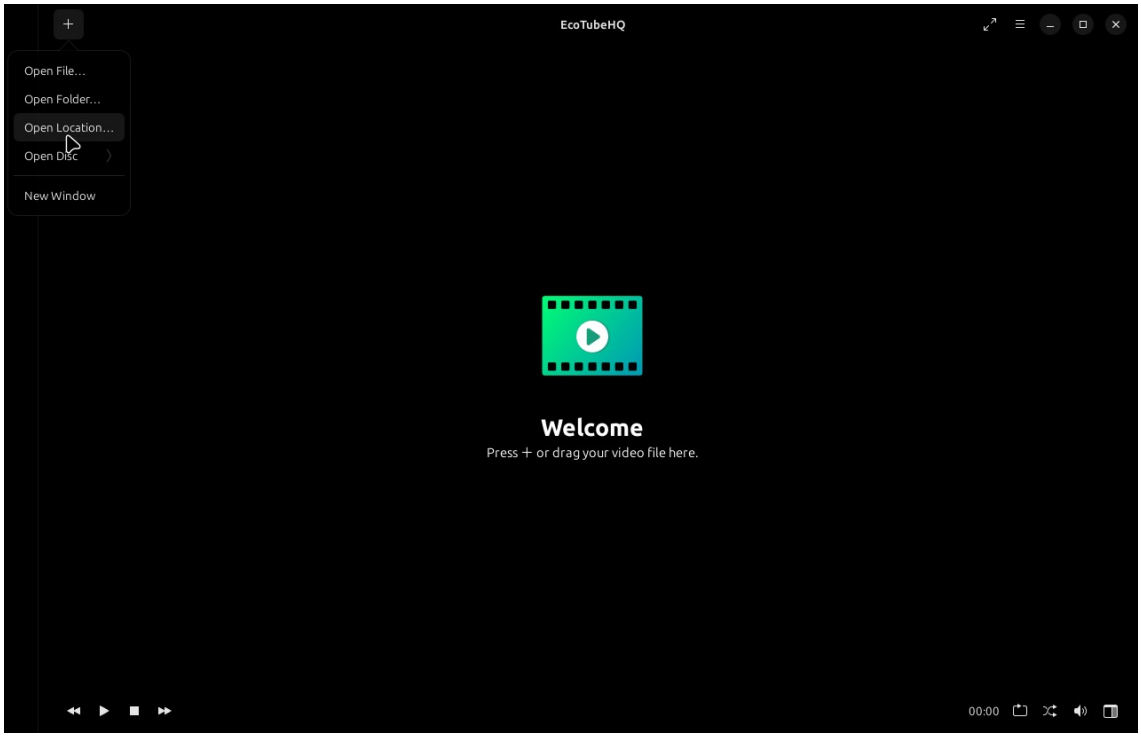
Transmitting data from YouTube's servers to your PC consumes energy, and in the aforementioned example the conventional buffer wasted 81 seconds more A/V data than the minimal buffer of EcoTubeHQ.

81 seconds of A/V data may seem like a tiny amount, but when you consider over 1 billion hours of YouTube videos are watched every day it is clear that the amount of data wasted every day is mind boggling.

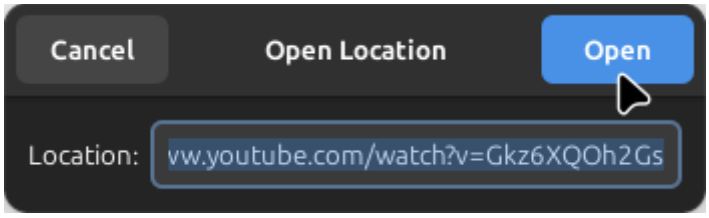
A nominal buffer also benefits systems operating under network bandwidth constraints – less data wasted = less bandwidth used.

How to use

To play a video from an online streaming service simply copy the video URL from the browser, click '+' at the top left of the player and select 'Open Location'…



...the player automatically places the copied video location into the 'Location' entry box…



...click the 'Open' button to initiate playback.

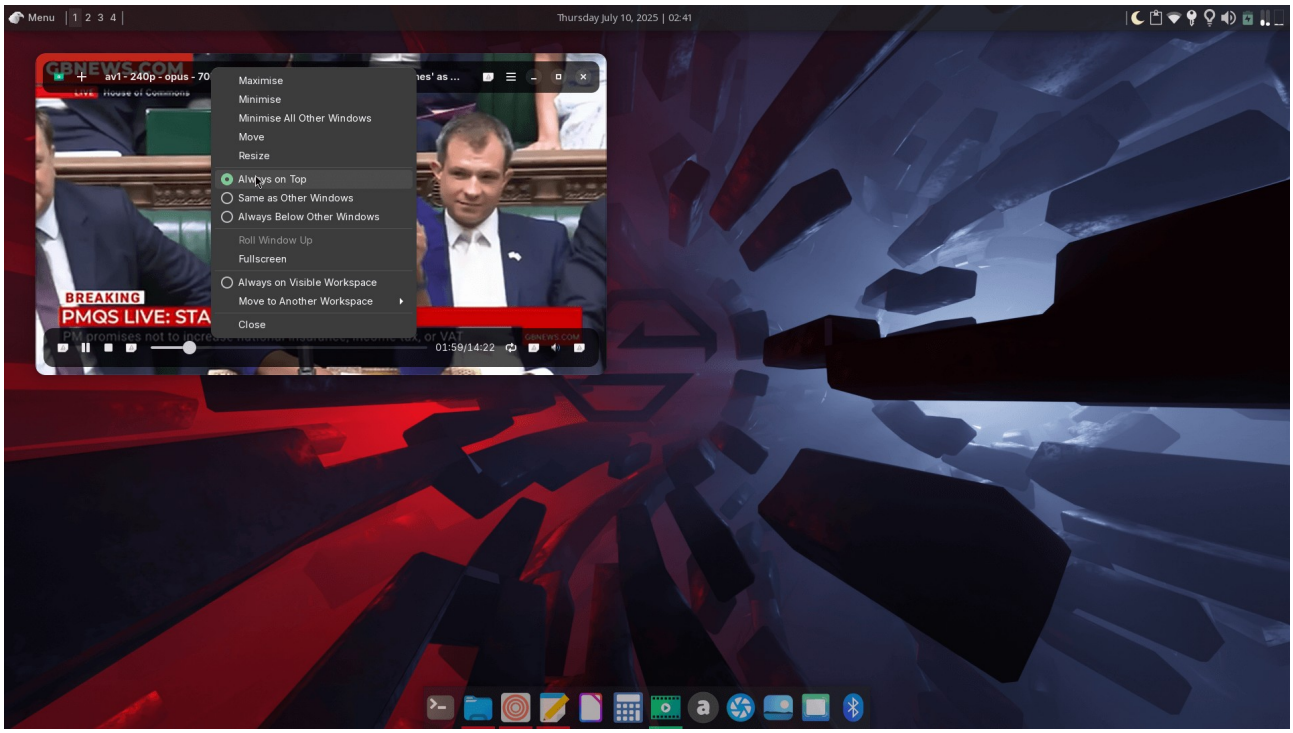
- Press the space bar or right click the mouse on the player screen to pause / play video playback.
- Rotate the mouse wheel on the playback window to turn the audio volume up / down – the volume will reset to normal when the player is restarted.
- Double click the player window during playback to play the video fullscreen.
- Press ‘Esc’ or double click on the player window during fullscreen playback to return the player to non-fullscreen playback.

Miniplayer Usage

If you are watching a YouTube video which does not require fullscreen or you want to minimize CPU usage during playback then set the player's video output resolution to 144p or 240p.

This automatically reduces the size of the player's non-fullscreen window, effectively providing you with a miniplayer. The size of 240p videos is shown in the screenshot below, and 144p videos are displayed in a smaller window.

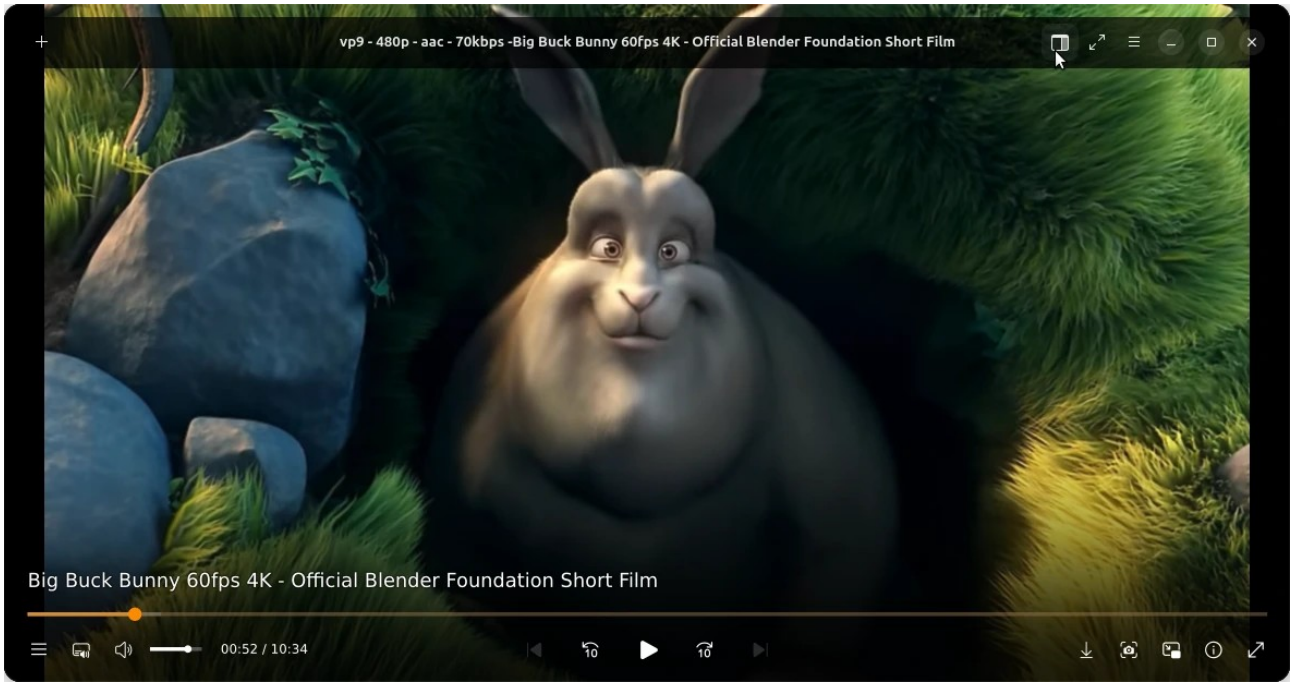
To keep the player on top of all other windows during playback right-click on the top bar and select 'Always on top' as shown below.



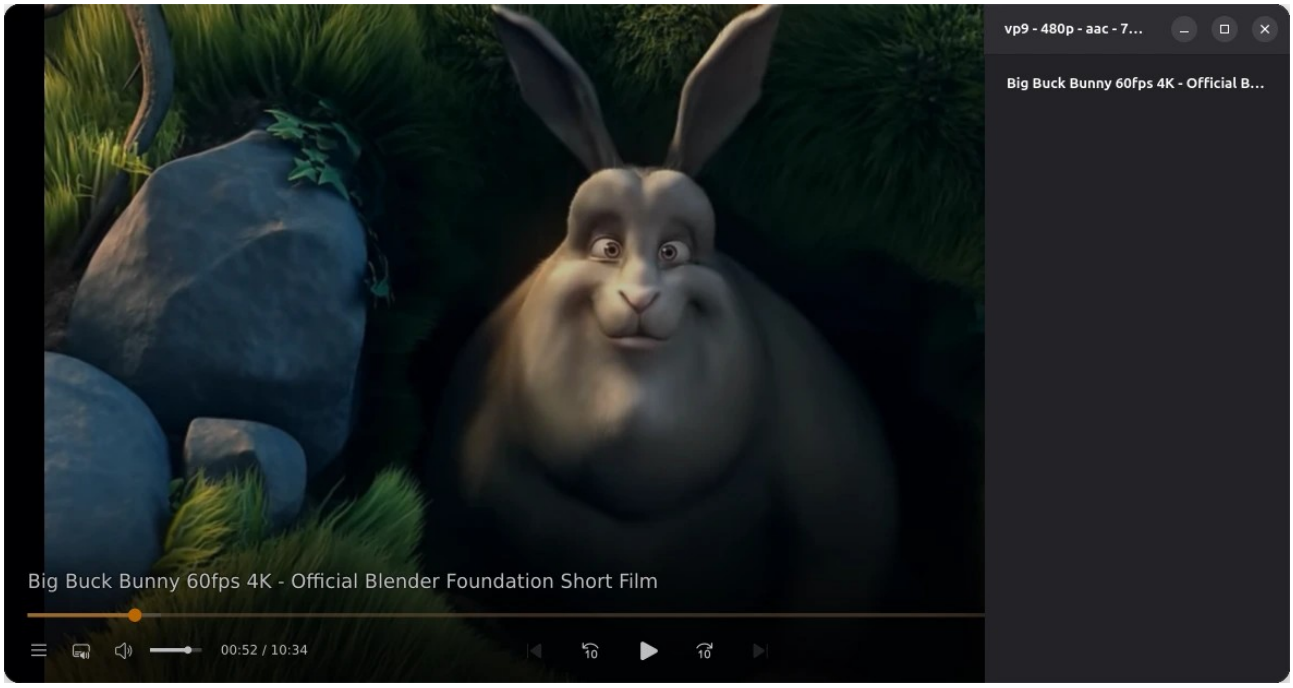
The miniplayer significantly reduces the amount of data used for videos of things such as podcasts where video quality is not paramount.

* To further reduce CPU usage set the Playback Type to **Powersave**

Playlists



To add a video to a playlist click on the top icon indicated above. When this is done the playlist opens.



To add videos from an external streaming service drag and drop the browser URL for the playlist onto the player’s playlist. Additional videos can be added prior to, and during playback. To start playback simply double-click on the required playlist video.

Screenshots

To take a screenshot of the on-screen picture click the button at the bottom of the player shown below…

…the image will be stored in the **Pictures** folder.

Audio and video information

To see information relating to the on-screen video click the information button shown below…

EcoTubeHQ Original Concept: Colin Bett

Coding and additional ideas: Sako Adams

Forked from the Celluloid Video Player

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Changelog

26.03.01

- Updated Preferences and removed HDR option
- Changed video upscale option to nearest neighbor
- Fixed format selection
- Changed app name in Start Menu / launcher

v26.02.01

- Updated preferences and removed deprecated features
- Fixed dark theme issue
- Fixed issue occurring on Wayland on Open Location option
- Auto update ffmpeg

v25.12.02

- Add option to set many seconds of audio/video to prefetch
- Added mpv reload script
- Update ecotube fidelityfx super resolution logo

v25.12.01

- Fixed thumbnail on sandbox
- Disabled video download
- Enhance interface
- Reorganized Auto mode

v25.11.01

- Enhance interface
- Added modern OSC with screenshot and stats
- Fix FSR when plugged in
- Add ‘auto’ mode in desktop instance

v25.10.2

- yt-dlp is auto-updated when a new release is available
- Added modern On Screen Control
- Local files show thumbnails on the seekbar
- Download the video
- Enhance UI

v25.10.1

- Added ‘auto mode’ in playback type
- Fix stream latency
- Enhance UI

v25.9.3

- Updated yt-dlp
- Bilinear upscaling is used for powersave mode
- Enhance UI

v25.9.2

- Fix codec selection in powersave mode
- Updated playlist mode
- Enhance UI

v25.9.1

- Updated playback options - Powersave and Quality
- Updated playlist mode
- Added HDR option in preference
- Enhance UI
- Fixed memory usage in startup

v25.8.2

- Added two playback options - Powersave and Quality
- Removed redundant codec selection
- Reorganized default playlist
- Included no https protocol in format selection
- Reduced network stream usage

v25.8.1

- Forced yt-dlp to use MPEG-DASH on some resolutions
- Reduce stream when paused
- Fixed resolution dropping
- Exclude HDR
- Add playback type

v25.7.2

- Forced mpv to avoid HLS - temporarily

v25.7.1

- Added Vulkan support
- Fixed video / audio download + improved video loading
- Fix format selection
- Updated caching system
- Fixed auto-resize player

v25.5.1

- Upgrade to gtk4.16 and libadwaita-1 1.6
- Enhance interface with playlist overlay
- New update notification
- New dialog for "How to use it"
- Minor playback fix

v25.3.3

- Buffering: prefetch is now limited to 10 seconds to minimize data used when videos are not played in their entirety.
- Hardware acceleration is automatically activated if a laptop is running on battery power.
- Auto update parsers.
- Updated the way default resolutions are selected.
- The preference interface has been updated.

V25.2.1

26 February 2025

Initial conversion from Celluloid.

Thanks to haggen88 for the icon update and albanobattistella for the patches,

14 August 2024
Initial commit to Github.